Operational Benefits of Standardization and Open Architectures

Semantic Technology

for

Intelligence, Defense, and Security Conference

STIDS 2026

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Operational Ontology Application: DoDAF Meta Model (DM2)

- Establish and define the **constrained vocabulary** for description and discourse about DoDAF models and their usage in the six core DoD processes
- Specify the **semantics and format for federated Enterprise Architecture (EA) data exchange** between architecture development and analysis tools and architecture databases across the DoD EA Community of Interest (COI) and with other authoritative data sources
- Support **discovery and understandability of EA data**
- **Improve EA data discovery** using DM2 categories of information
- **Improve understandability of EA data** using DM2's precise semantics augmented with linguistic traceability (aliases)
- Provide a **basis for semantic precision** in architectural descriptions to support heterogeneous architectural description integration and analysis in support of core process decision making

Foundation for DoDAF Formal Ontology

- The DoDAF Meta Model (DM2) is founded upon the **International Defence Enterprise Architecture Specification (IDEAS)**
  - Formal ontology foundation developed by the defense departments and ministries of the United States, United Kingdom, Canada, Australia, and Sweden in coordination the North Atlantic Treaty Organization (NATO)
- All DoDAF concepts and concept relationships inherit several **rigorously defined mathematical properties** from the IDEAS Foundation
- The **W3C Resource Description Framework (RDF) and Web Ontology Language (OWL)** are specified as the format to be used for data exchange

Purpose of the DoD Architectural Framework

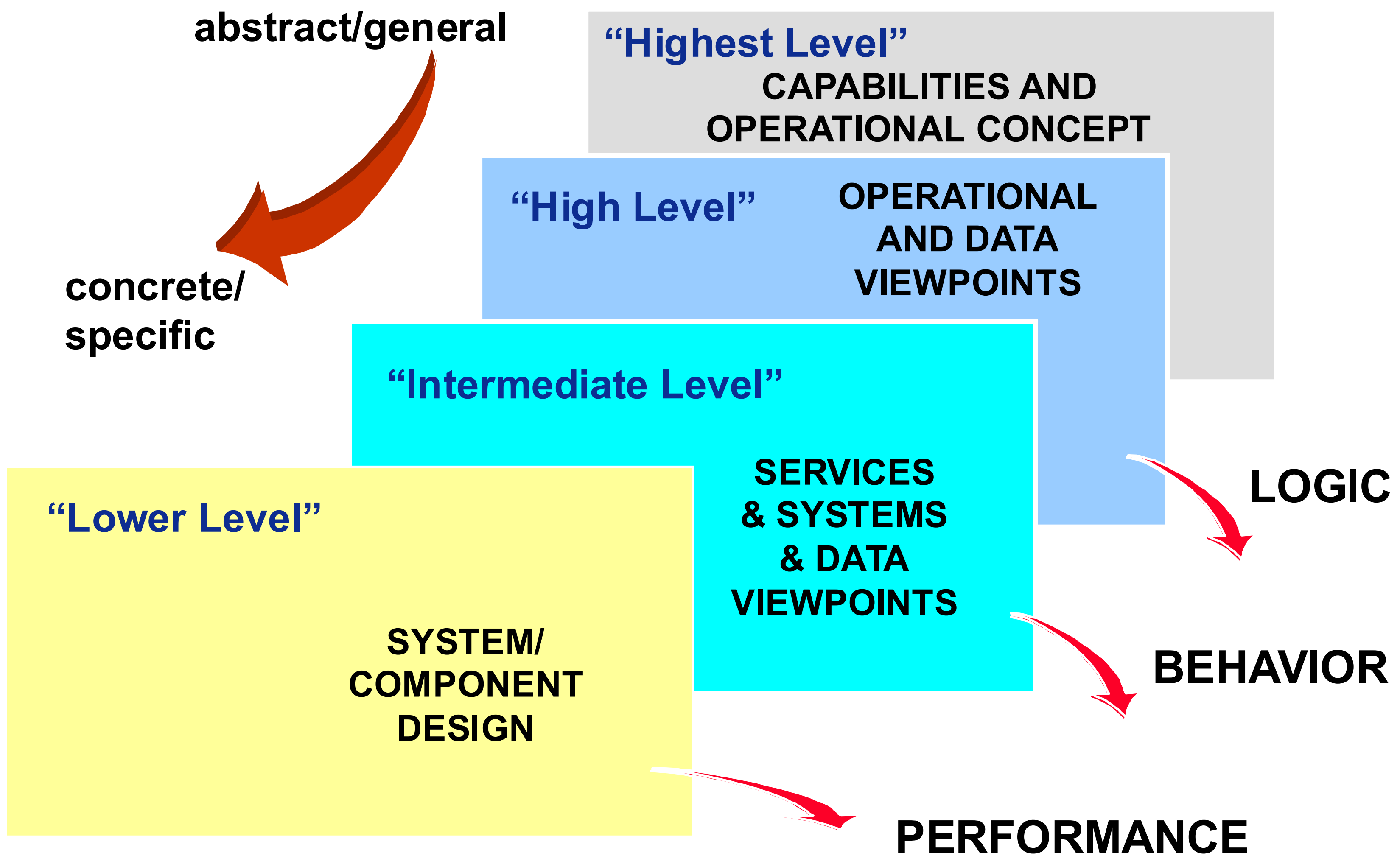
The purpose of DoDAF is to define concepts and models usable in DoD's six core processes:

- **Capabilities Integration and Development (formerly JCIDS)**
- **Planning, Programming, Budgeting, and Execution (PPBE)**
- **Acquisition System (DAS)**
- **Systems Engineering (SE)**
- **Operations Planning**
- **Capabilities Portfolio Management (CPM)**

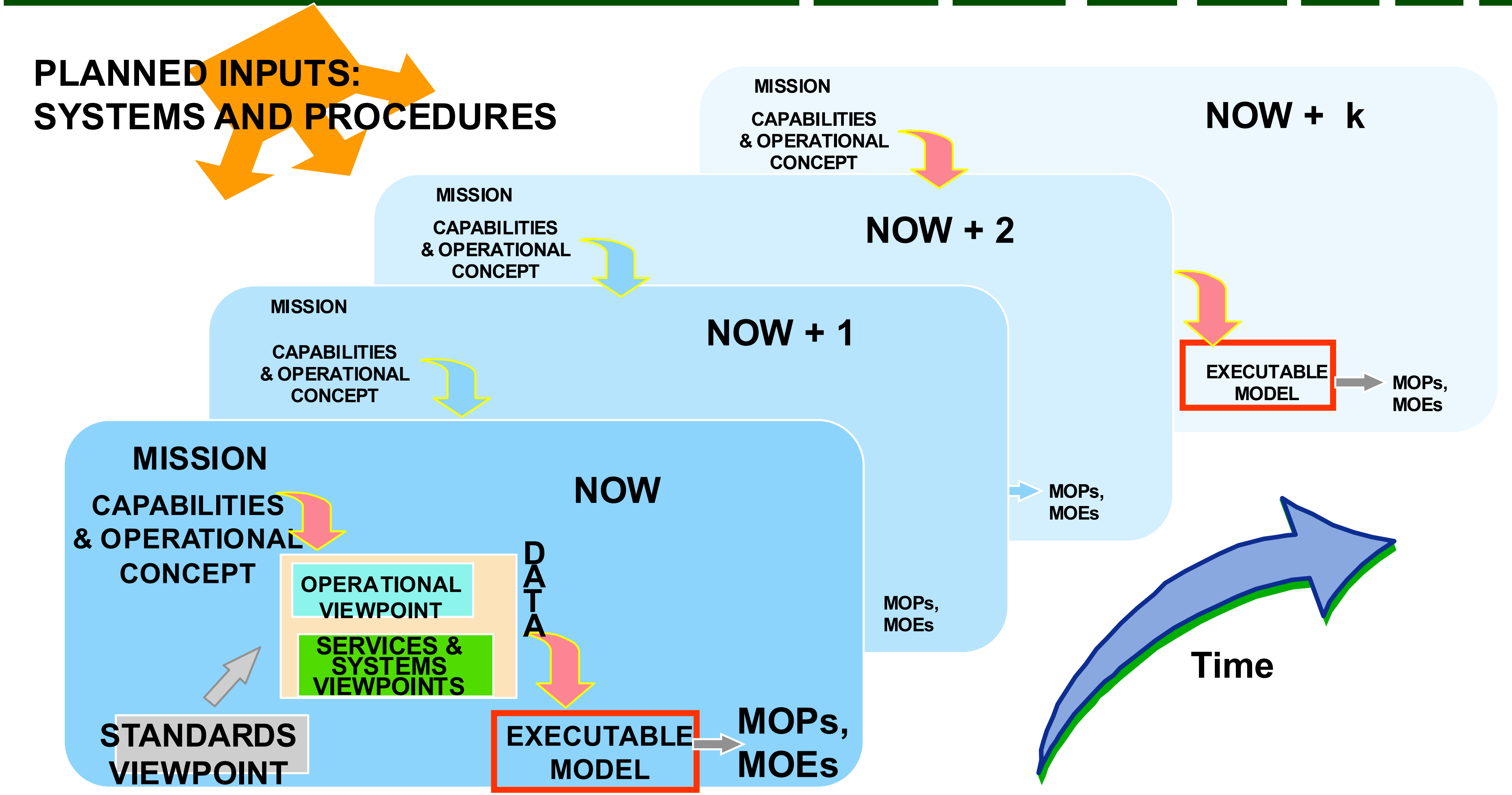
DoD Architecture Design Approach

- **The approach used for architecture design is one of:**
  - **Successive refinement:** from higher levels of abstraction (capabilities and operational concepts) to lower-level ones (systems architectures) all the way to component design
  - **Evolution:** Since architectures last for a long time, we need to consider not only the near future, but the far future
- **C2 processes evolve continuously and new systems/components enter while other ones are retired—this requires**
  - **Data interoperability – a non-trivial requirement**
  - **Collaboration**

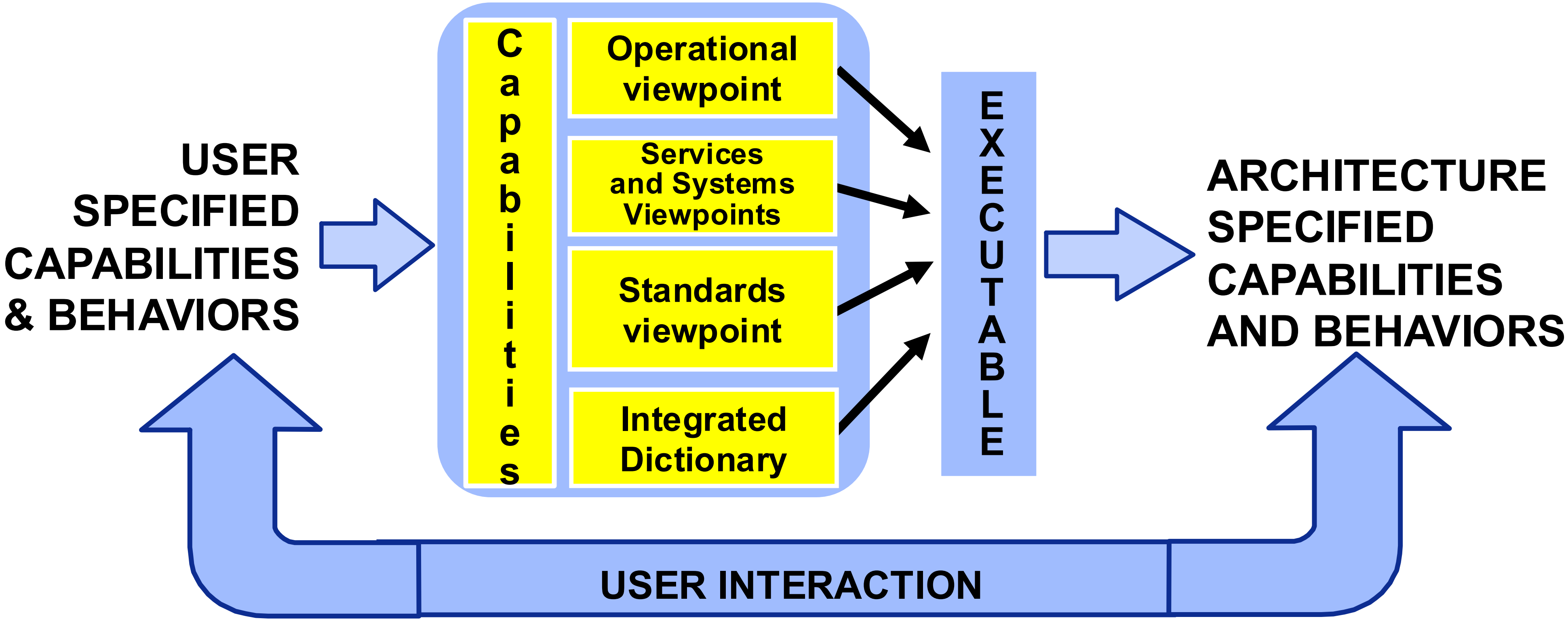
Process Of Refinement



Evolution in Time

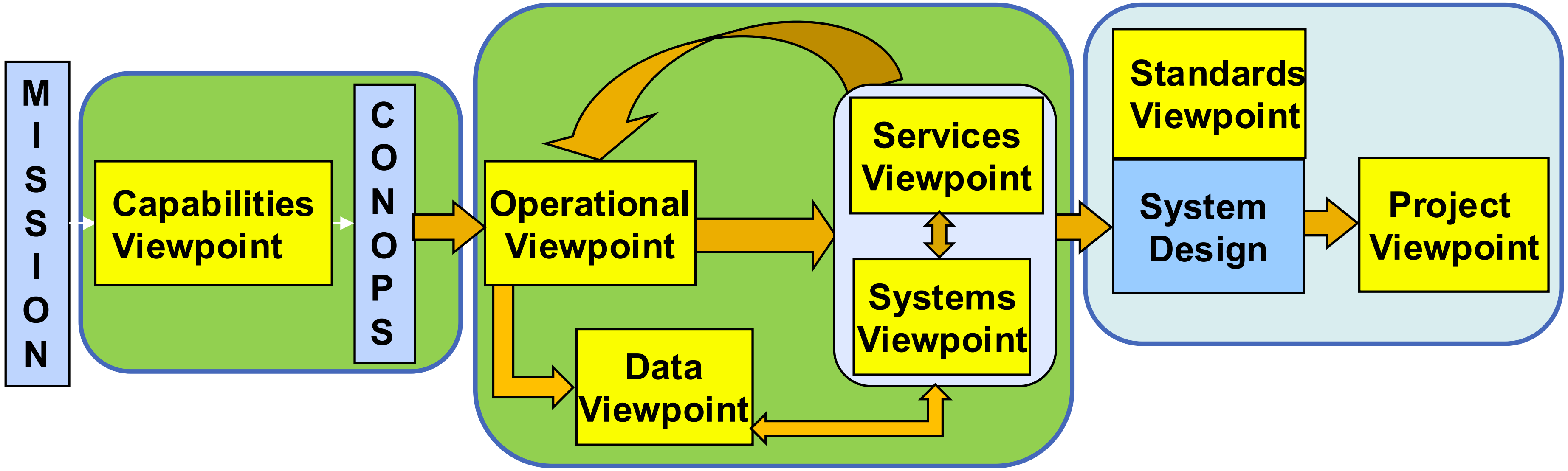


The Focus of the Design is on Capabilities



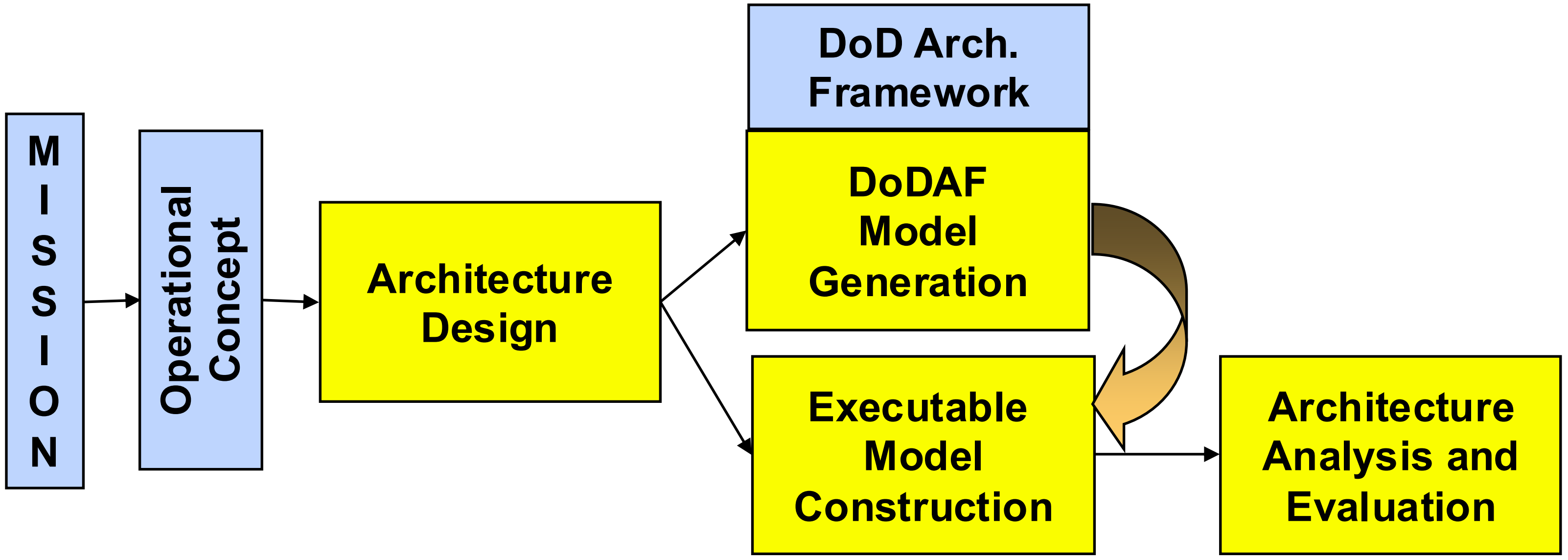
User behavioral and performance requirements vs. architecture enabled behavior and performance

The Basic Architecture-based Design Steps



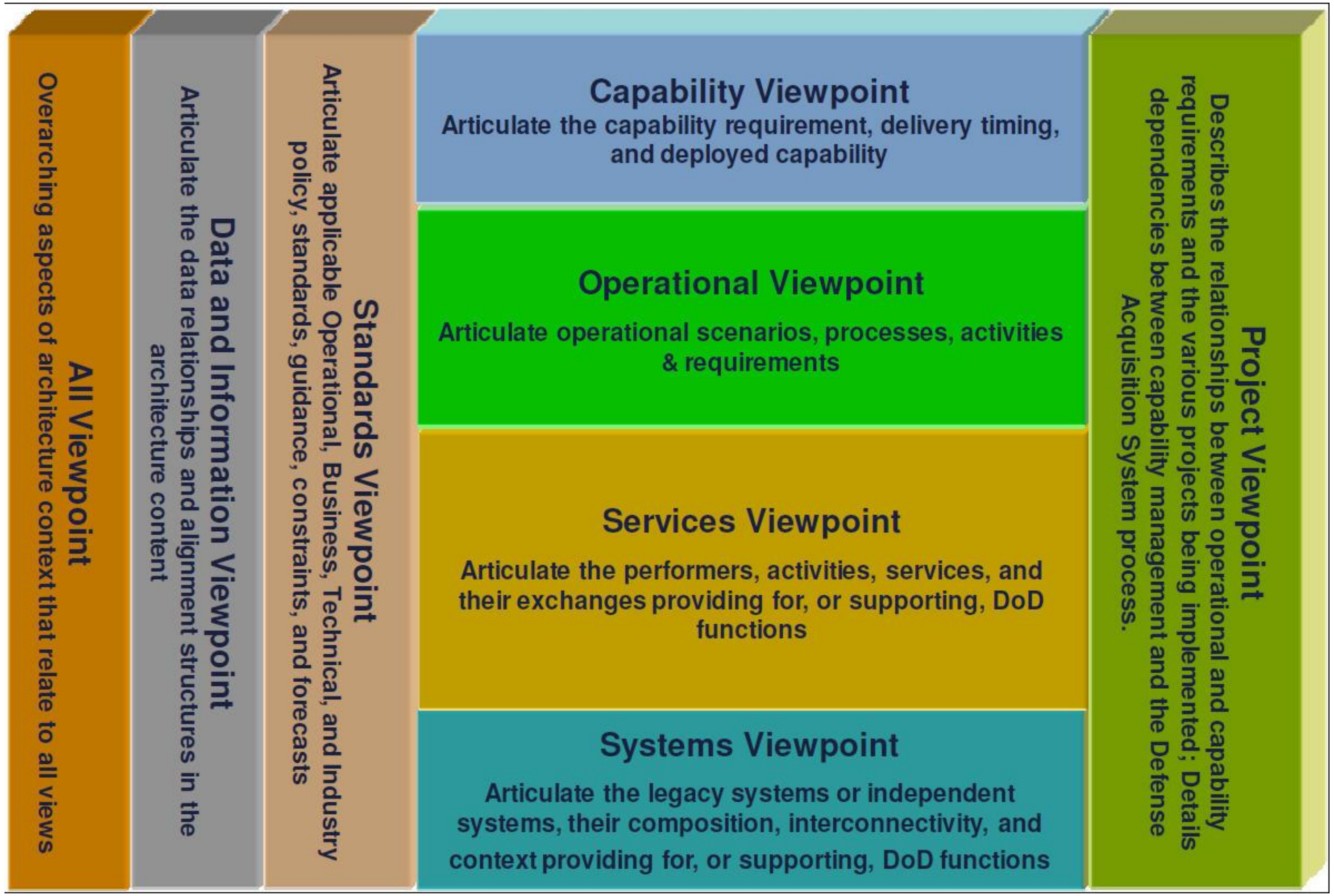
- Given the Command's Vision and Mission statement, we need to articulate a set of capabilities and the associated operational concept(s)
- The Operational view expresses the functional requirements that are then satisfied by construction in the services and systems viewpoints.
- The process is iterative – with iterations representing refinement levels until actual designs are obtained

The Overall Architecture Design Process



- Alternative approaches and tools can be used to design the DoD Architecture models and viewpoints – with standard ontologies, this is not an issue
- The executable model is used for behavior and performance evaluation
- Note: Architectural efforts are *goal* or *problem* based and not "architecture *model* based"

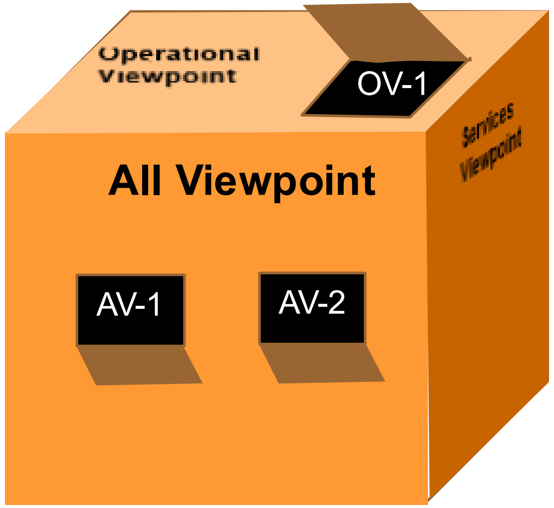
The DoDAF 2 Viewpoints

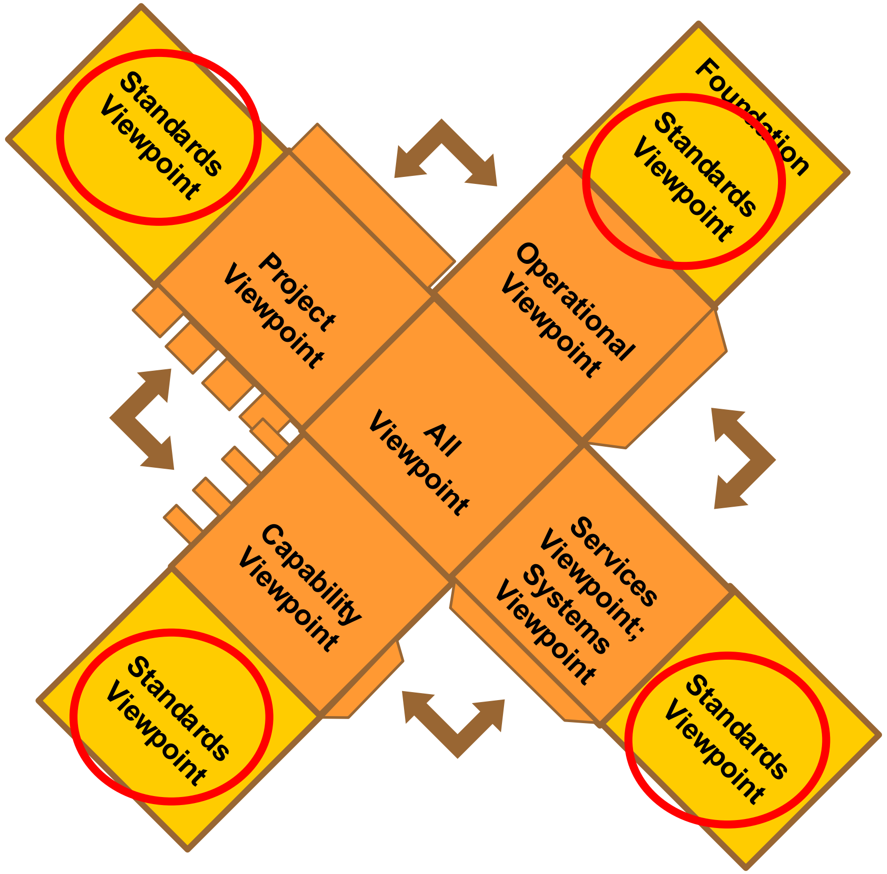


DoD Architectures are described in conformance with the DoD Architecture Framework 2.0

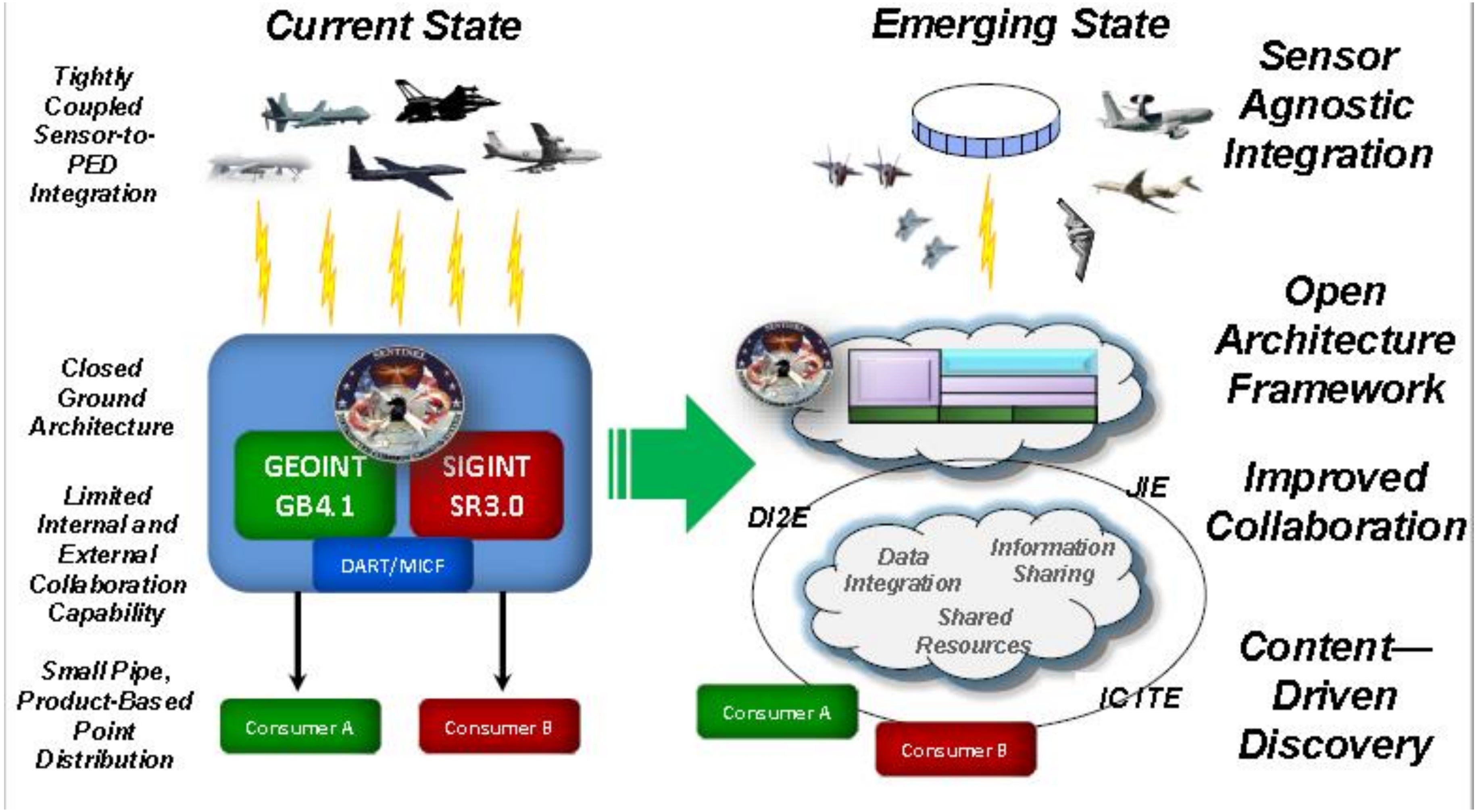
Relationships between DoDAF Viewpoints

Each viewpoint has an associated
Standards Viewpoint
appropriate to that viewpoint





Operational Example: Transforming AF DCGS



AF DCGS Focus Areas

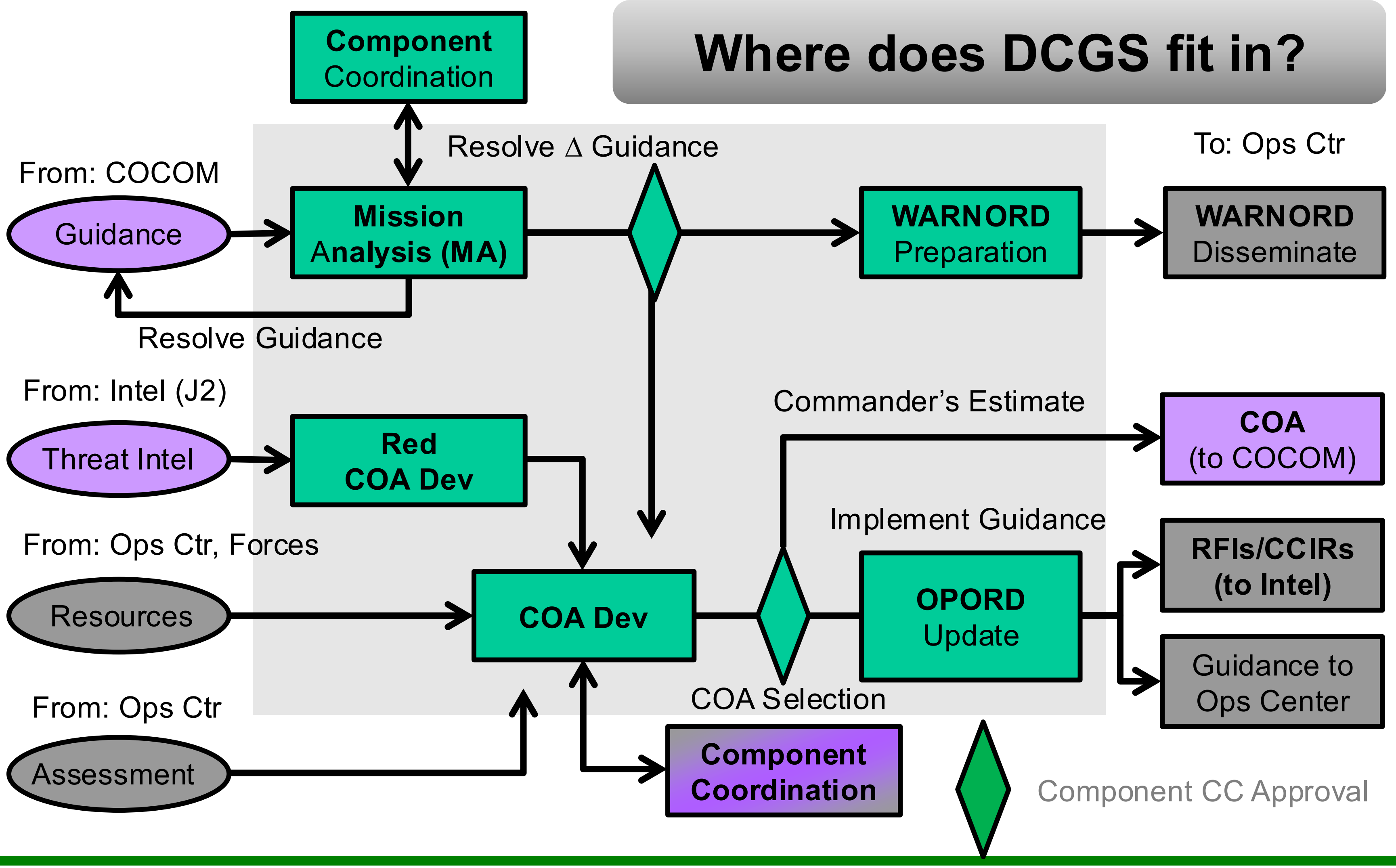
- Ability to Rapidly Integrate New Data Sources
- Improve Info Sharing, Integration & Interoperability
- Access to Multi-INT Analysis and Fusion Capabilities
- Ability to Rapidly Deliver Capability to Warfighter
- More Effective and Efficient (Reduce Cost of Ownership and Sustainment) Weapon System
- Robust, Cyber-Resilient Weapon System

Air Force DCGS (with CFACC)

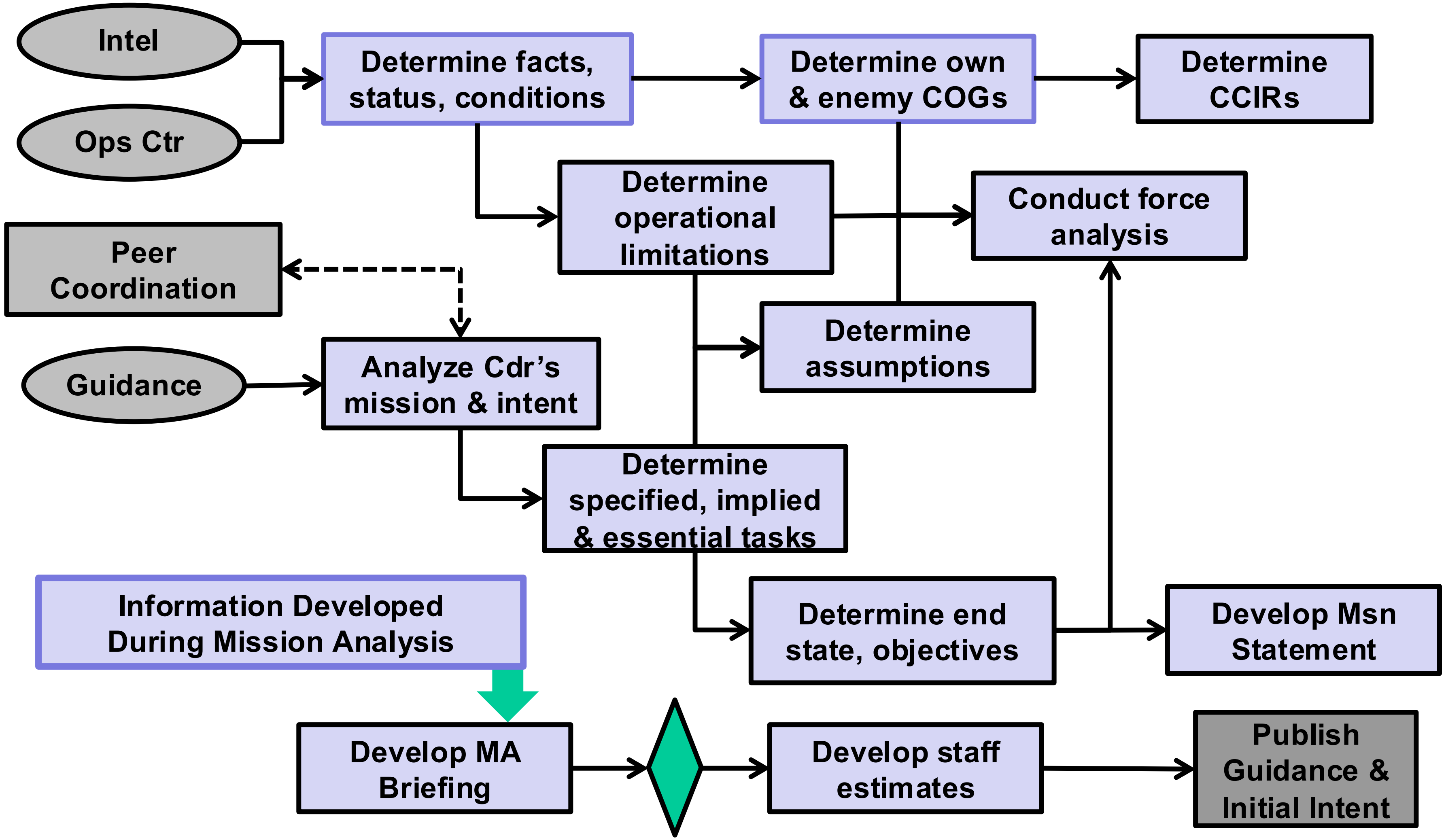
CFACC "User" Needs

- The Air Force uses AF DCGS to
  - Connect to the multi-Service DCGS Integration Backbone (DIB)
  - Manage requests for sensors
  - Process sensor data
  - Exploit sensor data from multiple sources
  - Disseminate intelligence products.
- To support CFACC intelligence needs, Air Force DCGS must be able to
  - conduct time-dominant and decision-quality analysis to optimize ISR operations,
  - produce timely assessments
  - enhance battlespace awareness and threat warnings

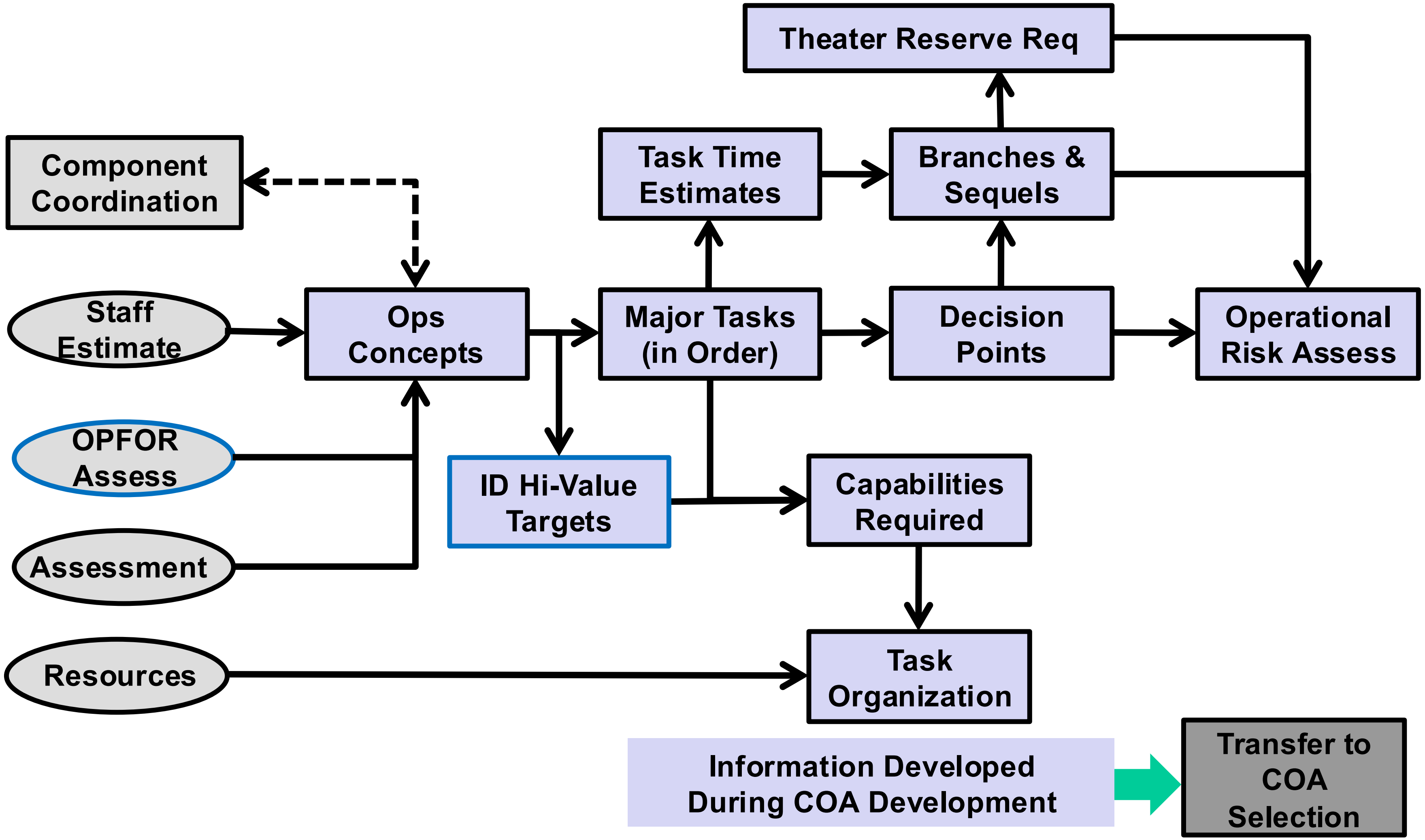
High-level CFACC Overview



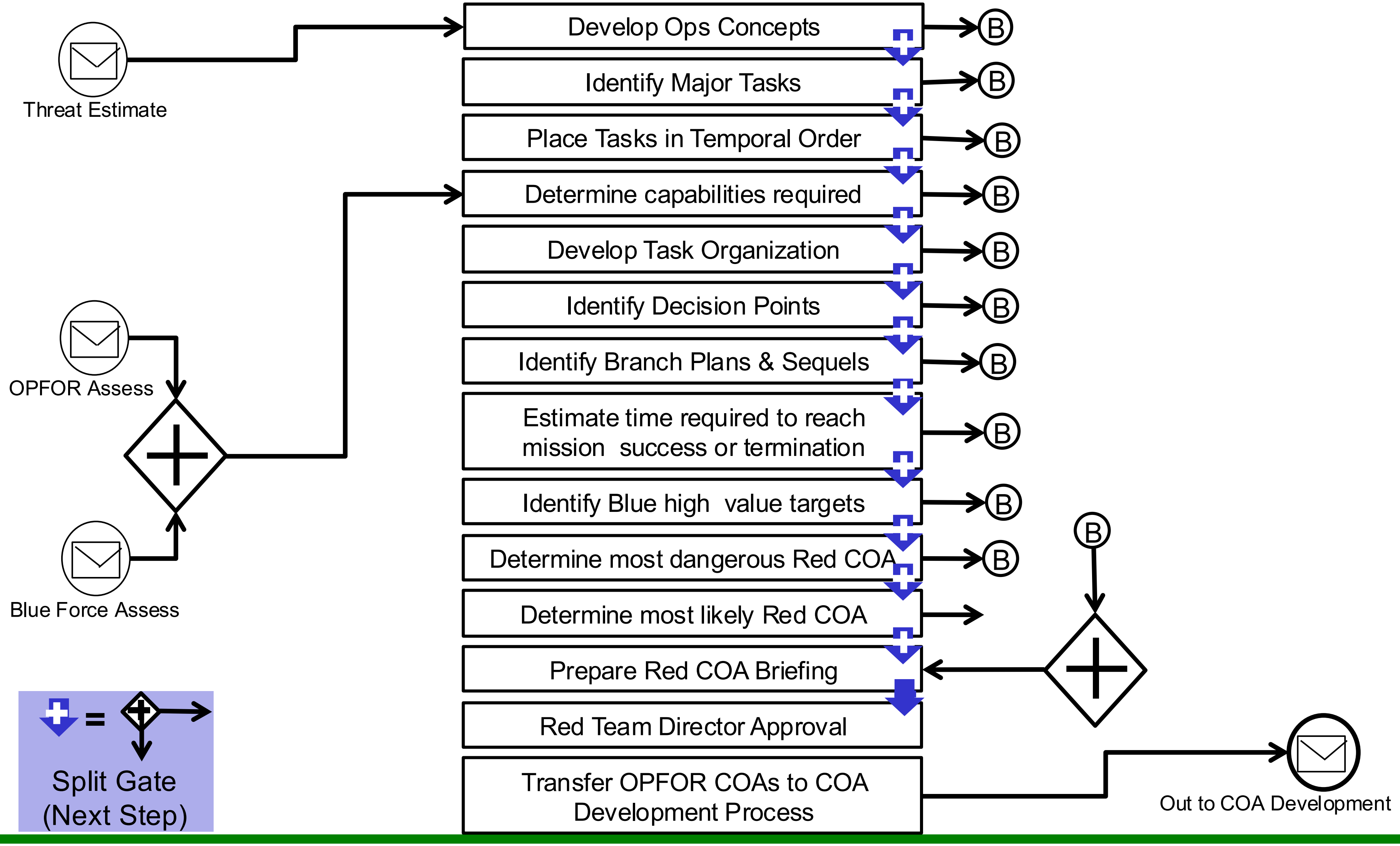
Mission Analysis (CFACC Staff)



CFACC Staff COA Development



Opposition Force (OPFOR) Business Process Model

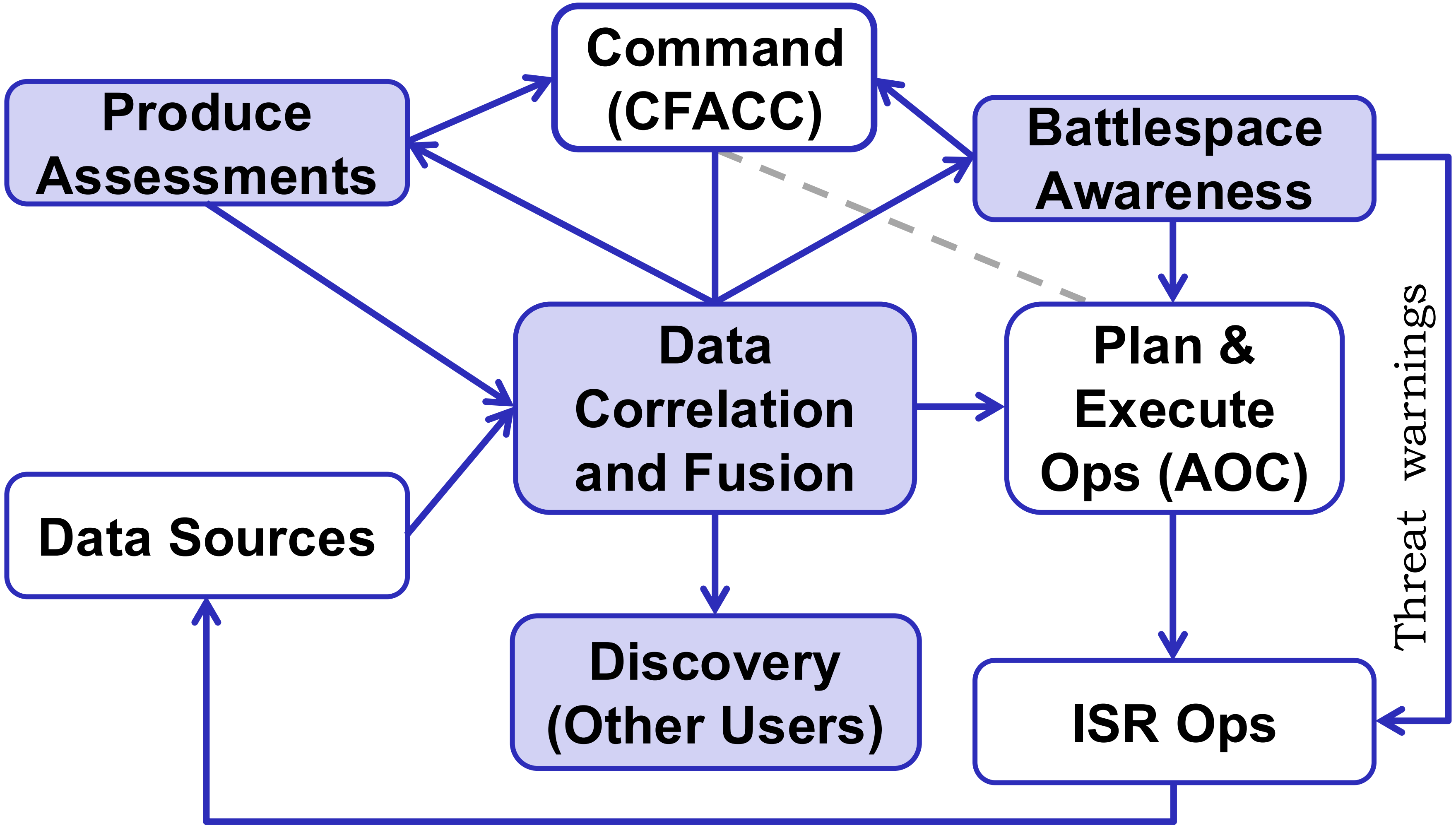


Other DCGS Actors (Partial List)

Many different needs, all to support CFACC

- ISR Group Commander
- Component Numbered Air Force
- Multi-Source Analyst
- Geospatial Intelligence (GEOINT) Fusion Analyst
- Signals Intelligence (SIGINT) Fusion Analyst
- Supported Unit
- Correlation Analyst (CAN)
- Ground Liaison Officer (GLO)
- Distributed Mission Site (DMS) Mission Planning Cell
- Liaison Officers
- Mission Operations Commander
- Mission Crew
- Exploitation Crew
- Air Operations Center (AOC)
- Fusion Lead Msn Planning Cell
- Platform Mission Planning Cell
- Multi-Source Analyst
- Geospatial Intelligence (GEOINT) Fusion Analyst
- Signals Intelligence (SIGINT) Fusion Analyst (SFA)
- Fusion Lead Requirements Mgr
- Analytic Partner
- Supporting Intelligence Unit
- Requirement Submitter
- Analysis and Reporting Cell
- Imagery Mission Supervisor
- Ground Mission Supervisor
- Multi-Source Mission Supervisor

Combined DCGS and AOC Support to CFACC



Multi-domain C2 Problem Statement

- Staring with a clean sheet, develop a vision for Air Force Command and Control that adopts an **integrated approach for air, space, and cyberspace domains (as well as other domains: land, maritime, …) and is applicable at all levels of command.**
- Develop a high-level architecture that **incorporates evolving and future technologies** to provide value to the Air Force unconstrained by current processes and procedures.
- Outline a process for transitioning from the current evolutionary development of C2 and supporting C4ISR systems towards a **"To Be" architecture** at some future point in time (e.g., 2035)
- **Note: The Vision architecture is not the same as the "To Be" or "Target" architecture** that defines an implementation at some future point in time that is subject to structural and resource constraints

What is MDC2?

- **MDC2 is a collaborative multi-domain Command and Control process where all entities have the ability to share information and collaborate securely subject to classification constraints.**

MDC2 Vision

- The Air Force MDC2 should enable the timely delivery of integrated multi-domain options to the commander.
- The Air Force MDC2 should enable the development and execution of
  - integrated multi-domain, cross-regional, unified action (joint, coalition, whole of government and other partners) operations
  - under contested conditions (physical, spectrum, and information)
  - at all levels of command
    - National (POTUS/National Command Authorities)
    - Strategic (CCDRs)
    - Operational (JTF and Component Commanders) and
    - Tactical (entities controlling tactical-level operations)
- The Air Force MDC2 process should be scalable, resilient, and leverage advances in technology, personnel development, and command, control, and collaboration processes

Domain List

CSAF Domain List

- Air
- Space
- Cyberspace
- Land
- Sea
- Undersea

"Domain-like" Environments

- Information
- Spectrum
- Cognitive
- Social
- Financial
- Legal

Air Force Roles Requiring Multi-domain C2

- **Defend the Homeland**
- **Maintain global and regional stability (deterrence, assurance, escalation control, crisis management, and crisis response)**
- **Empower US and partner instruments of power to achieve strategic objectives**
- **Empower indigenous and partner military forces to prevent or favorably terminate regional conflicts**
- **Maintain cooperative relationships with international partners and open dialogue with US and partner competitors**
- **Fight the Nation's wars as part of the Joint Force**

Air Force Functions

- Domain (air and space) control
- Global precision strike
- Intelligence, Surveillance, Reconnaissance
- Global Mobility
- Global command, control, and battle management

Additional Commander MD-C2 Roles

- JCS Service Chief: National Unified Action (National Command)
- International Air Chief: International Unified Action
- JTF Commander: (Operational Command)
- JF Component Command (Operational Command)
- Multi-domain Tactical Ops Director (Control, Battle management)

MDC2 Usage

Situation	MDC2 Usage
Stable competition	Situation Understanding
Disturbance to Stability	Crisis Management
Crisis Situation or Contest Short of Armed Conflict	Crisis Response
Non-nuclear Conflict	Escalation Control Favorable Conflict Termination
Potential Nuclear-related Action	Competitor Deterrence Assurance Escalation control Favorable Crisis Termination

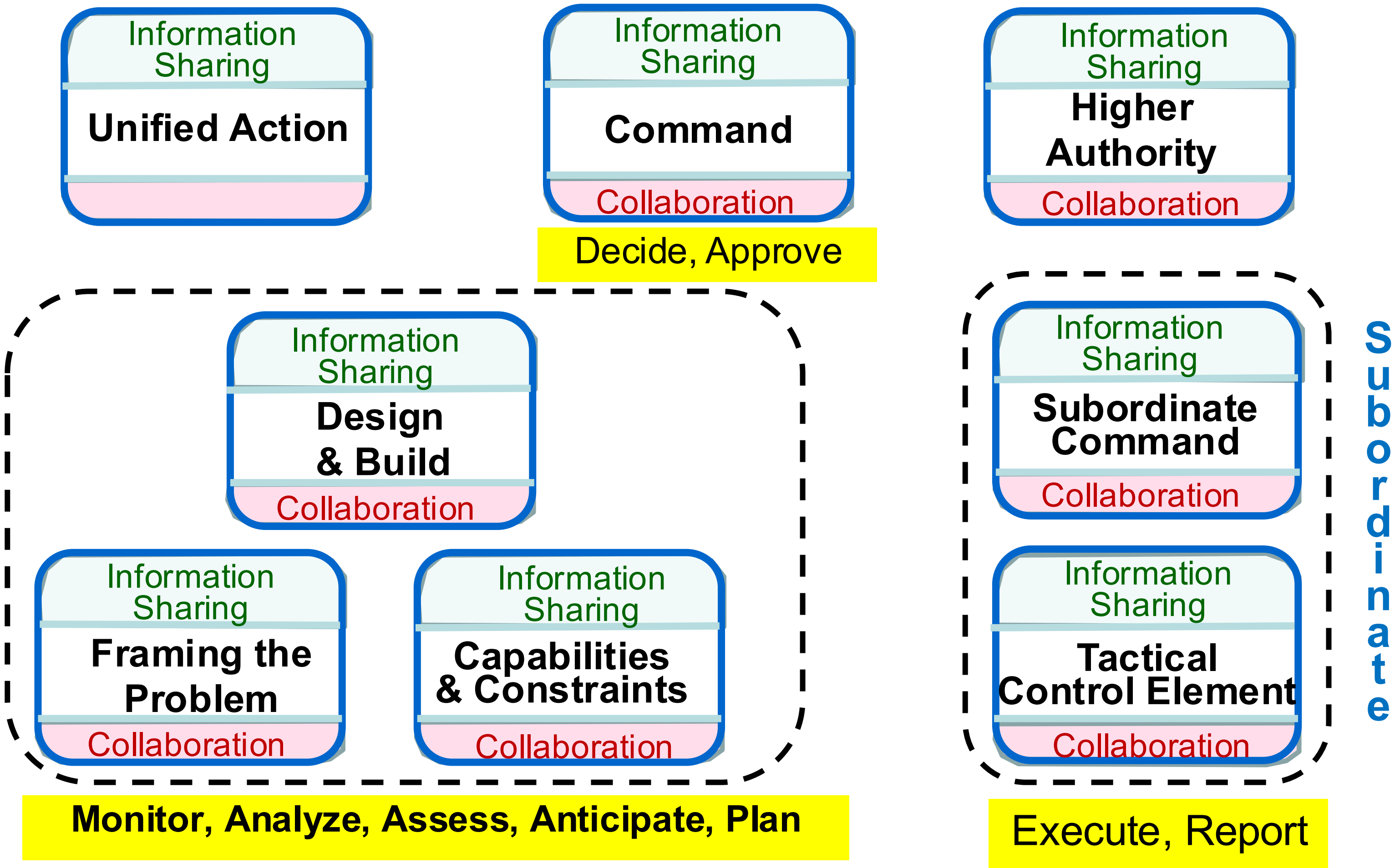
Relationships: Mission, Roles, Functions

Mission: Defend the United States and its global interests

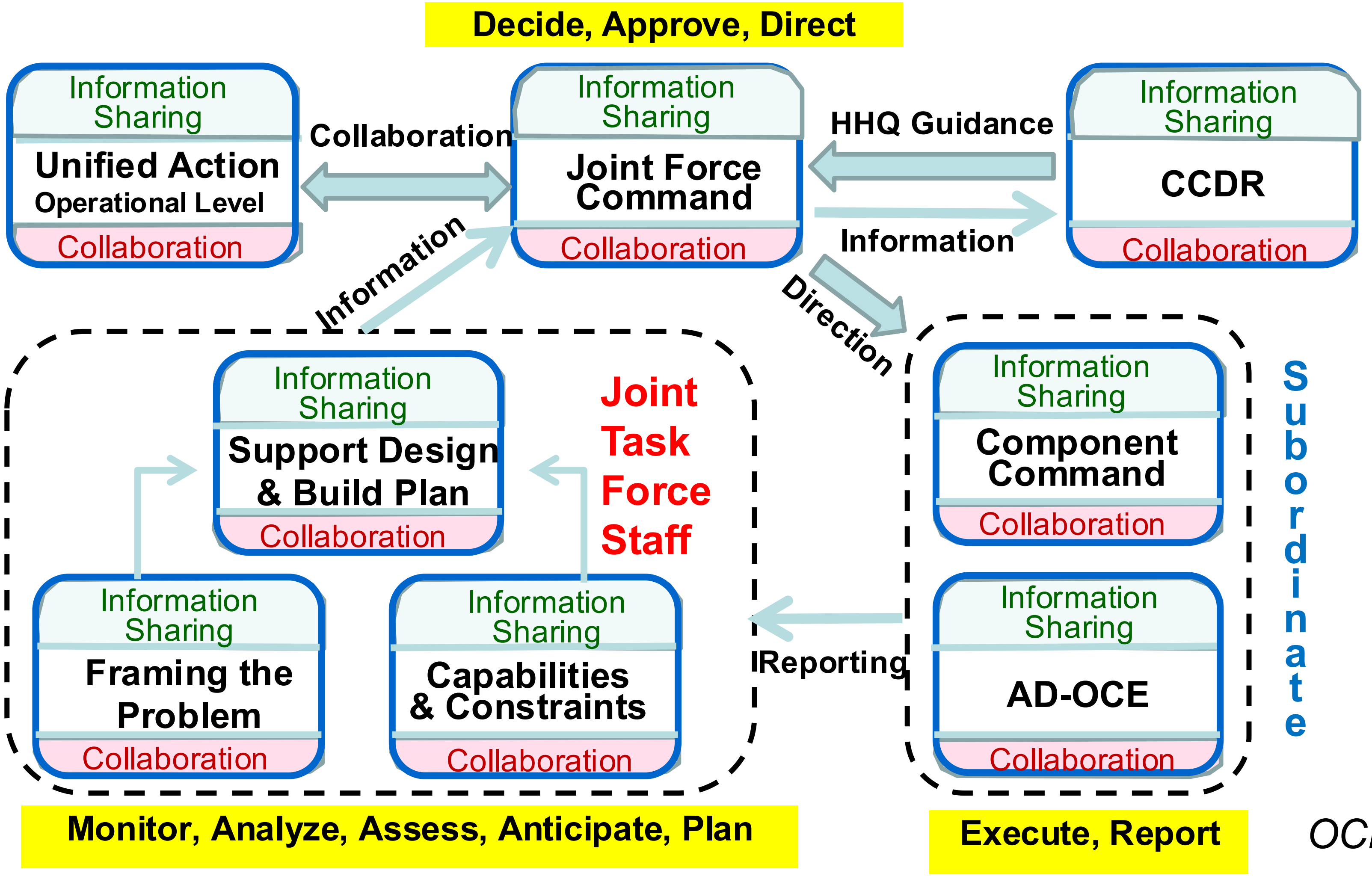
- Role: Defend the Homeland
- Role: Global & Regional Stability
- Role: Empower US and partner instruments of power
- Role: Empower partner military forces
- Role: Cooperative relationships
- Role: Conduct joint military operations

- Function: Domain Control
- Function: Global Precision Strike
- Function: Global ISR
- Function: Global C2 (& BM)
- Function: Global Mobility

The 8 Generic COIs for MDC2



Joint Force CMD Key Processes (C/JADC2)



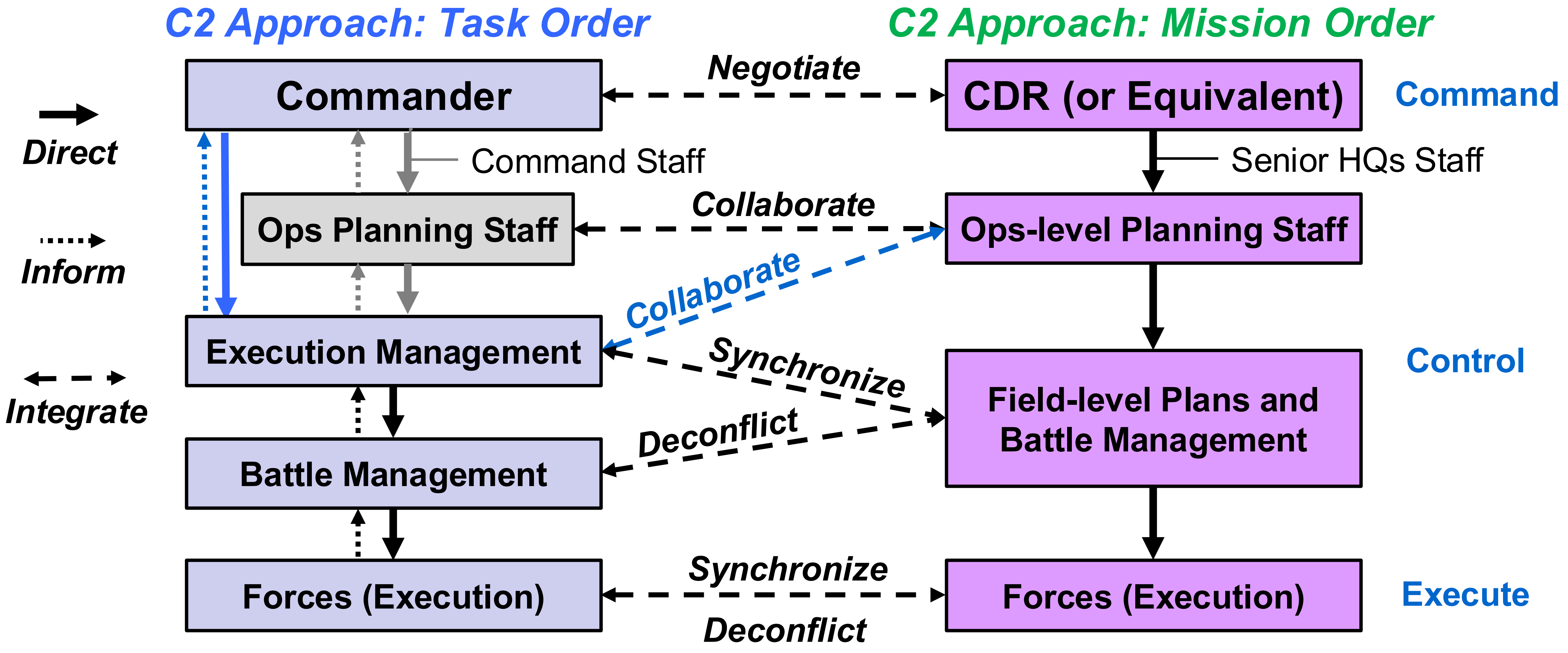
OCE: Ops Ctr Element

Four Levels of Command

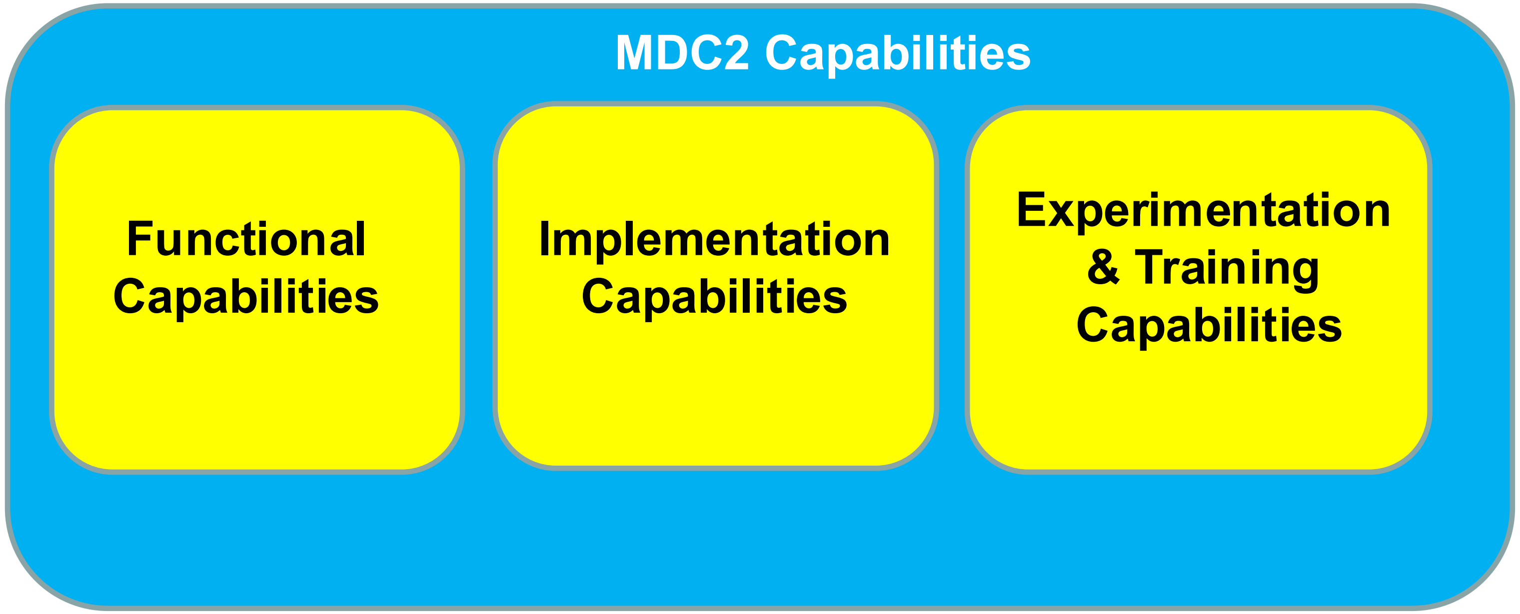
Community of Interest	National Command	Joint Forces Command	Component Command	Tactical Force Control
Command	POTUS (or National Command Authority)	Joint Forces Command	Joint Component Command	MD-OCE Director
Unified Action	Unified Action (Strategic Level)	Unified Action (Operational Level)	Unified Action (Operational Level)	Unified Action (Tactical Level)
Higher Authority	N/A (but responsive to Congress)	National Level	Joint Forces Command	Component Command
Design & Build	Joint Staff: Design & Build	Joint Forces Staff: Design & Build	Component Staff: Design & Build	N/A: Done at Component Level Allocate and Task
Frame the Problem	Joint Staff: Frame the Problem	Joint Forces Staff: Frame the Problem	Component Staff: Frame the Problem	MD-OCE Staff: Frame the Problem
Capabilities and Constraints	Joint Staff: Capabilities & Constraints	Joint Forces Staff: Capabilities & Constraints	Component Staff: Capabilities & Constraints	MD-OCE Staff: Capabilities & Constraints
Subordinate Command	Combatant Command	Component Command	MD-OCE/AFFOR	Subordinate Battle Management Entities
Tactical	Joint Chief Service Staffs	MD-OCE/AFFOR	Field Units	Tactical Forces

MD-OCE: Multi-Domain Operational Control Element

Different Approaches to Joint Command, Control, and Execution



MDC2 Capabilities Taxonomy



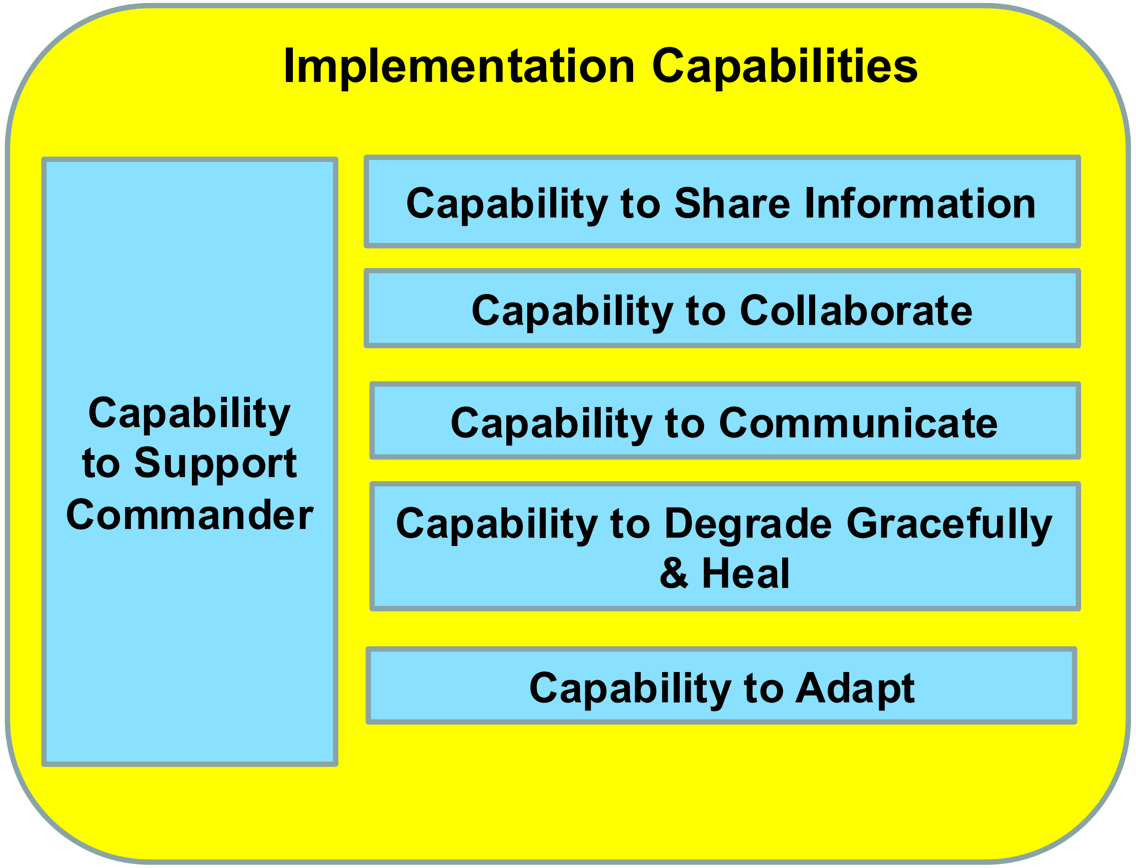
MDC2 Capabilities Taxonomy - 1

Functional Capabilities

Capability to integrate information from all Domains

- Capability to Frame Problem
- Capability to Design & Build
- Capability to Monitor/ Assess
- Capability to Direct
- Capability to Execute/ Report
- Capability to Exercise/Wargame
- Capability to Manage Risk
- Capability to Manage Execution

MDC2 Capabilities Taxonomy -2



MDC2 Capabilities Taxonomy - 3

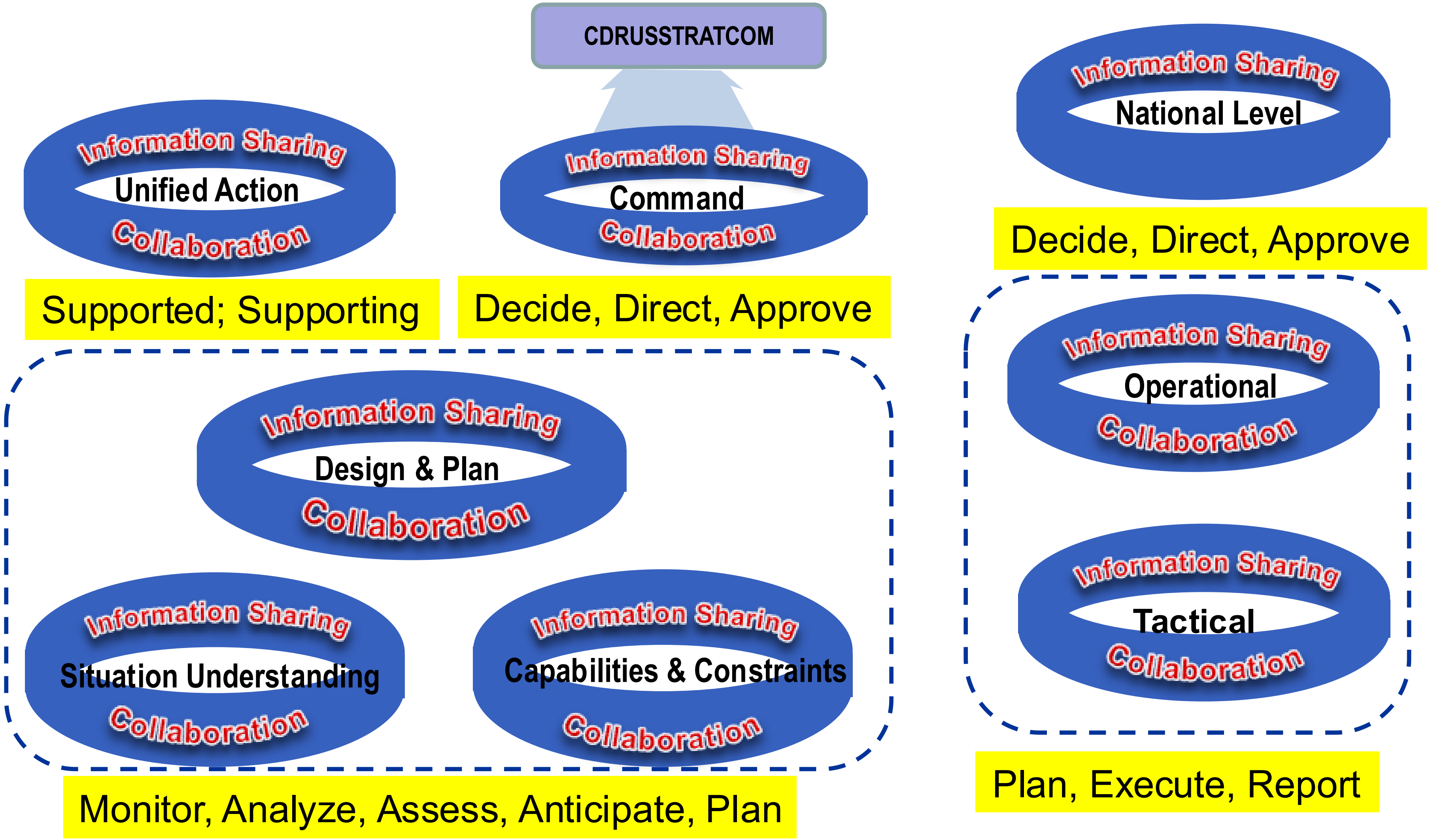
Experimentation and Training Capabilities

- Capability to conduct Experimentation Campaigns on the MDC2 process
- Capability Train MDC2 Operators/Staff
- Capability to Test and Evaluate C3 Systems and Services

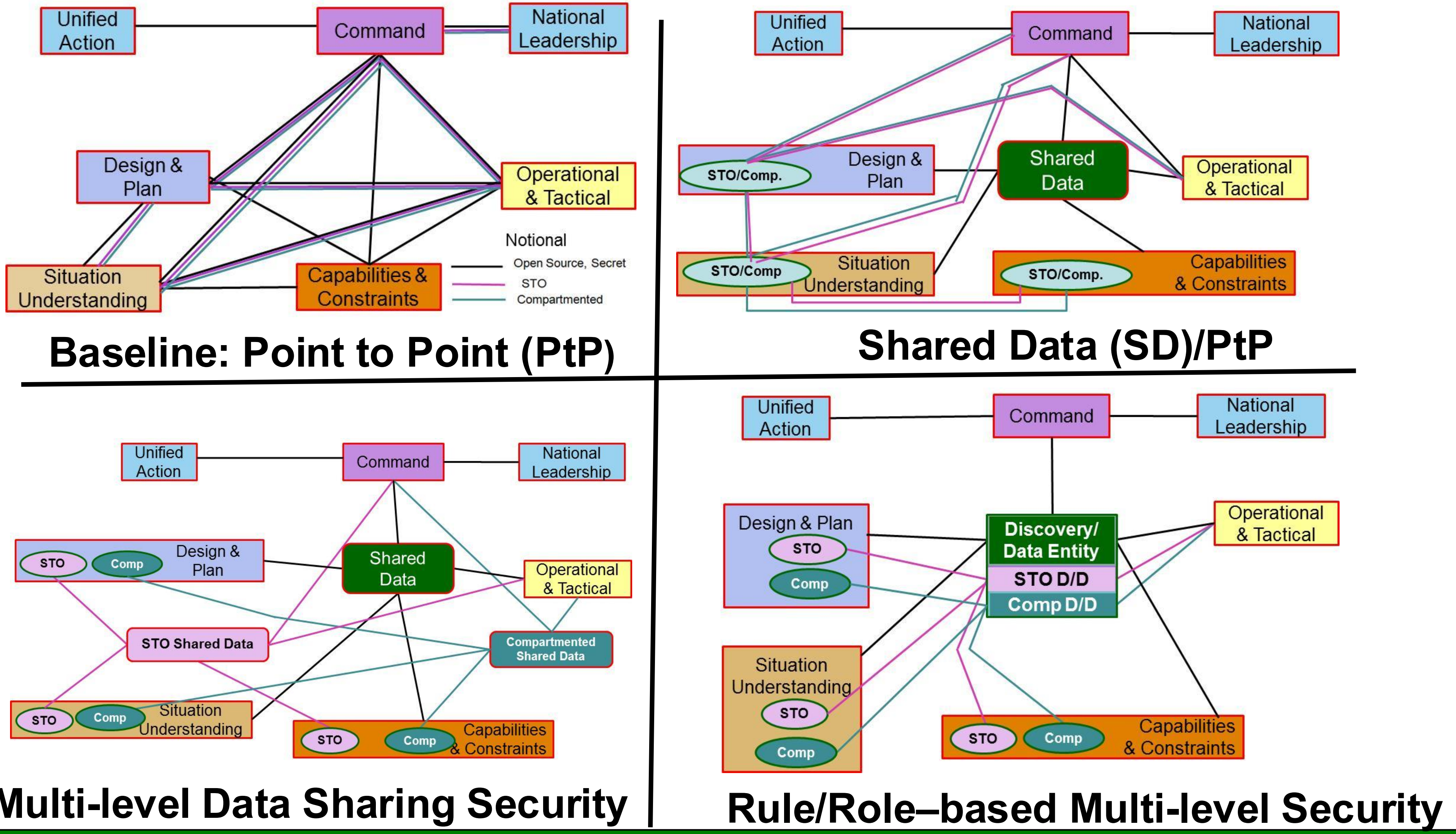
STRATCOM Integrated C2 Challenge

- The Command embarked on an architecture effort to define and implement an architecture for Integrated Command and Control (IC2)
- STRATCOM faced a three-tiered problem:
  - Define the set of capabilities that will enable the command the execute all its missions in an integrated manner
  - Ensure that Command and Control can be conducted in a distributed and distributive manner
  - Actualize the Command’s CONOPS to address the strategic, operational and tactical levels of operation across the command’s (then) three Lines of Operation (Global Strike, Space, Cyber)

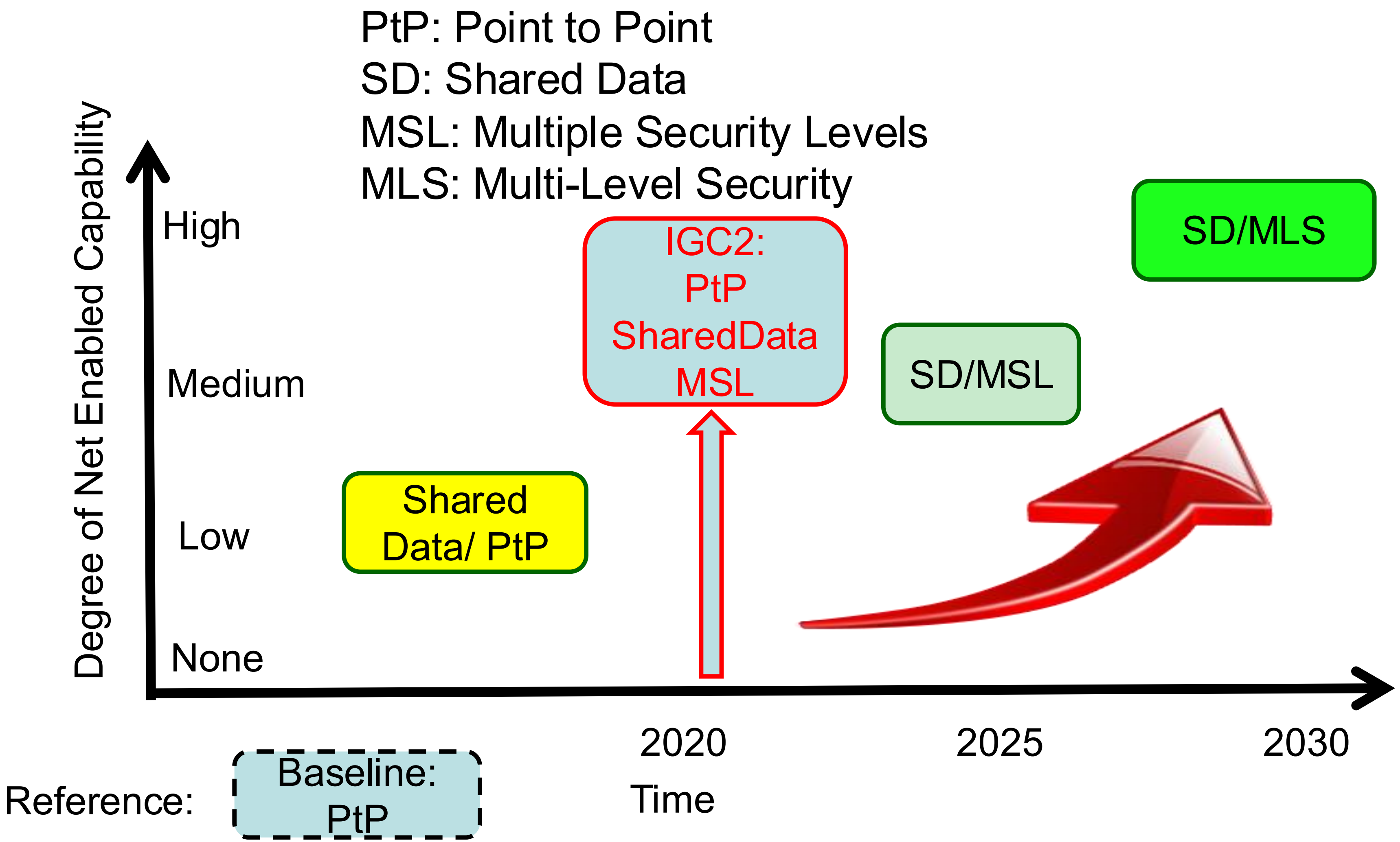
Communities of Interest



Architectural Approaches



C2 Architecture Evolution



Comprehensive C2 Architecture Objectives

Implementation of Comprehensive C2 Architecture supports DoD efforts to

- **Integrate and synchronize DoD-wide communications and infrastructure programs and efforts affecting JADC2, NC3, and National Leader C2**
- **Inform design, architecture, interoperability standards, capability development, and sustainment of critical command, control, and communications for joint, combined, and multi-agency operations across the competition-conflict spectrum of operations**
- **Identify opportunities to leverage emerging data, analytics, and machine intelligence/learning capabilities to increase decision advantage**

Comprehensive C2 Challenge

Comprehensive Command and Control (CC2) is defined as the comprehensive capabilities to command and control the entire spectrum of operations – *nuclear and non-nuclear*, military and non-military, U.S. and coalition – *from peacetime cooperation through competition, crisis, and conflict*, in all threat environments to include trans- and post-nuclear employment, aligned from the national through tactical levels of decision-making. (Working Definition)

Note: Comprehensive C2 elements are National Leader Command and Control, Nuclear C2, and Combined/Joint All Domain C2

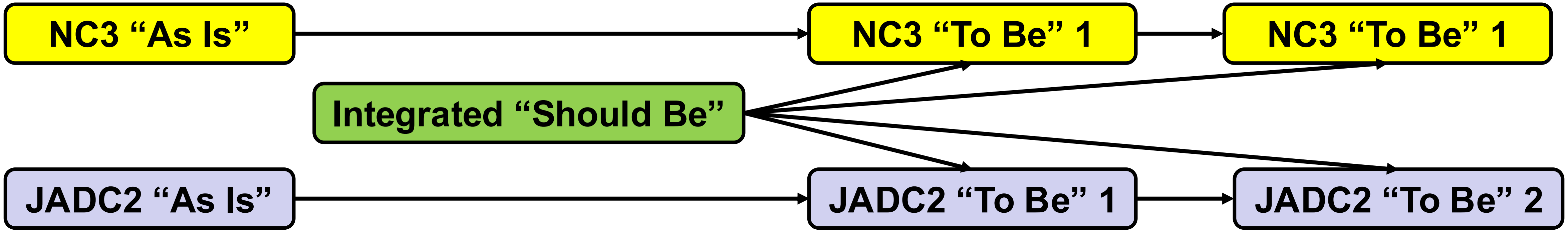
C/JADC2-related "Working" Definitions

Combined/Joint All Domain C2 (C/JADC2) is defined as the military capabilities to command and control *U.S. and international partner military operations* in all threat environments, aligned at the strategic through tactical levels of decision-making, across the *competition-conflict spectrum of operations*. (Working definition)

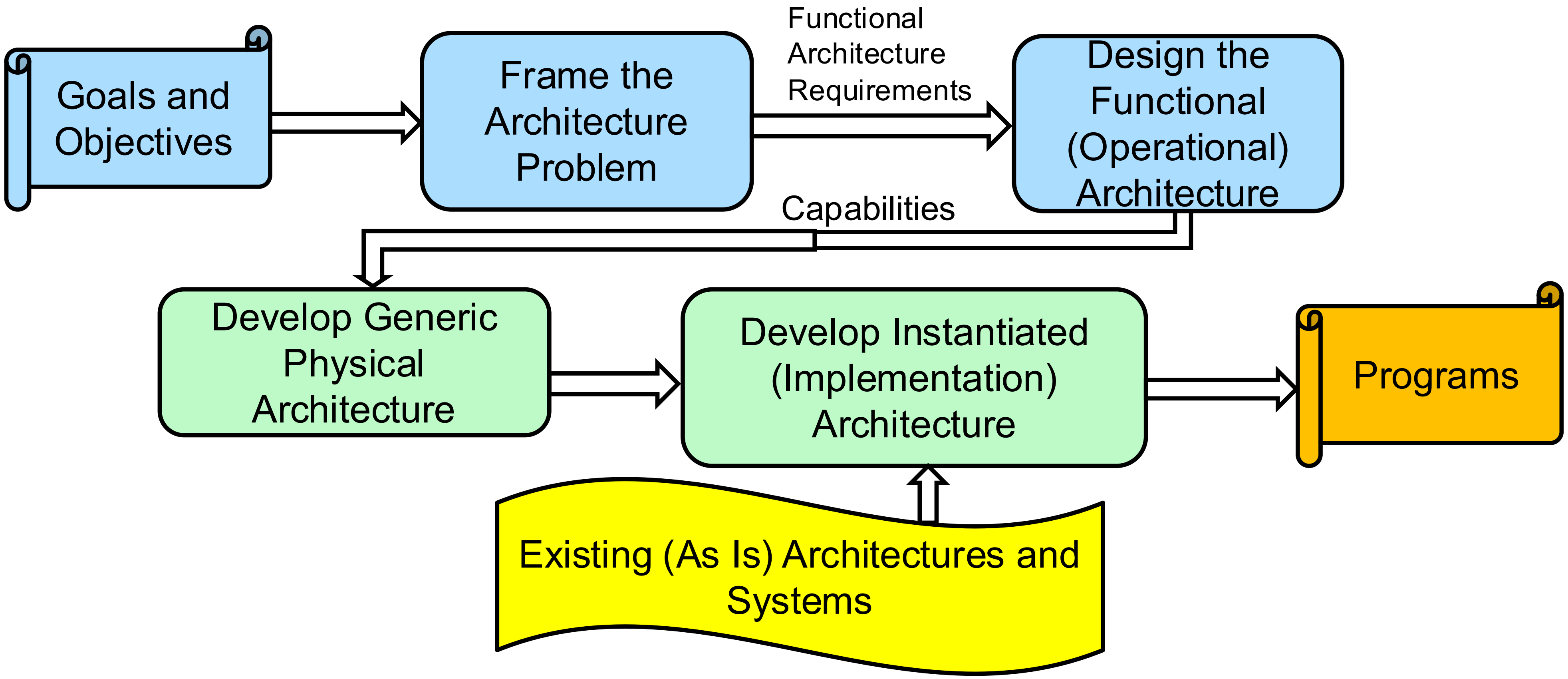
All Domain Effects Execution (ADE2) is defined as the warfighting capability to *sense, make sense, and act at all phases of competition and conflict*, across all domains, and with partners, to create *decisive advantage* through capabilities to find, fix, track, target, engage, and assess at the speed of relevance. (Working definition)

Comprehensive C2 Architecture Relationships

- NC3 “As Is” Architecture
- C/JADC2 “As Is” Architecture
- NC3 “To Be” Architectures (NC3 Next): Program Driven
- C/JADC2 “To Be” Architectures: Program Driven
- Integrated NC3/NC2-C/JADC2 “Should Be” Architecture
  - Phase 1: Frame the Integrated Architecture Problem
  - Purpose is to inform NC3 and C/JADC2 “To Be” Architectures

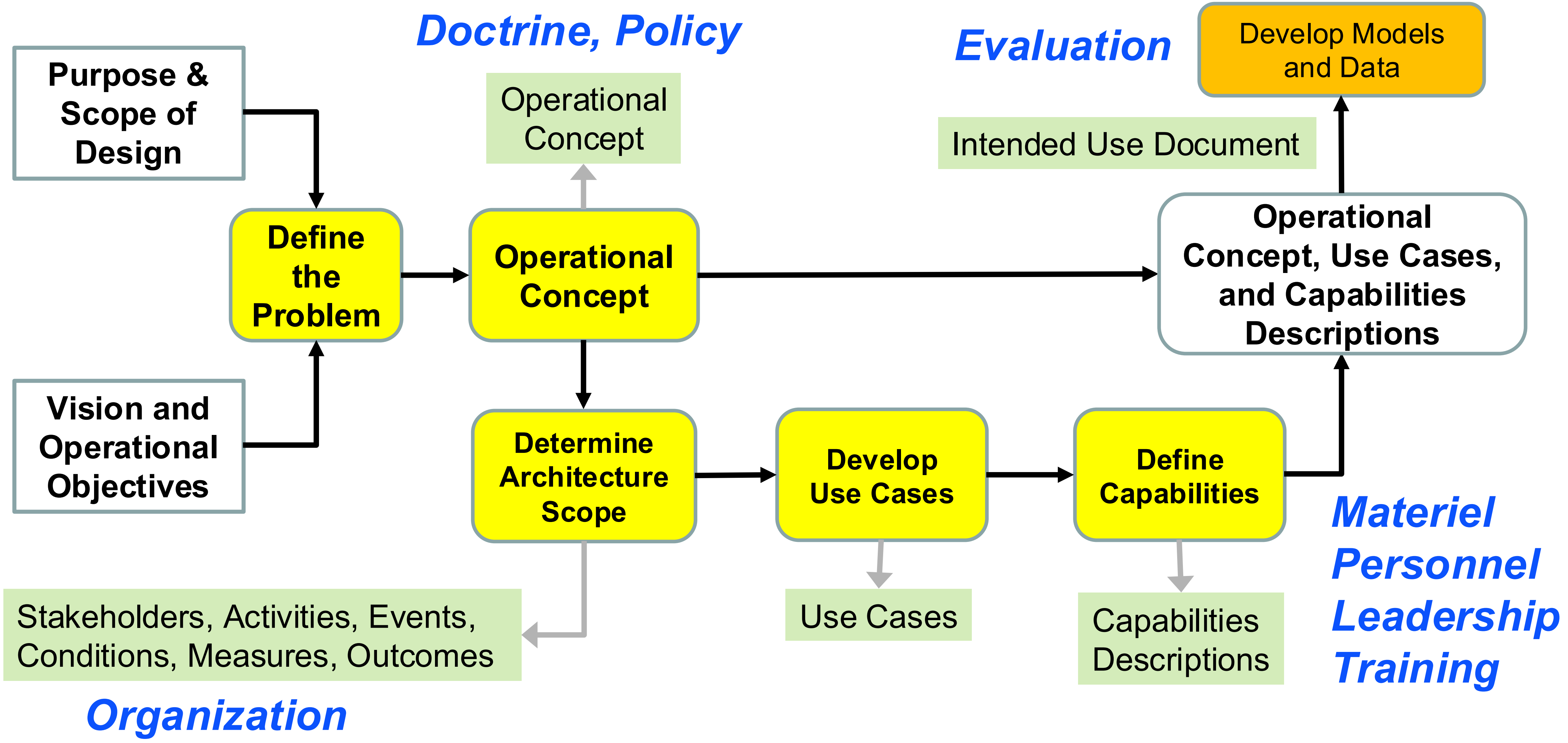


“Clean Sheet” Architectures Overview

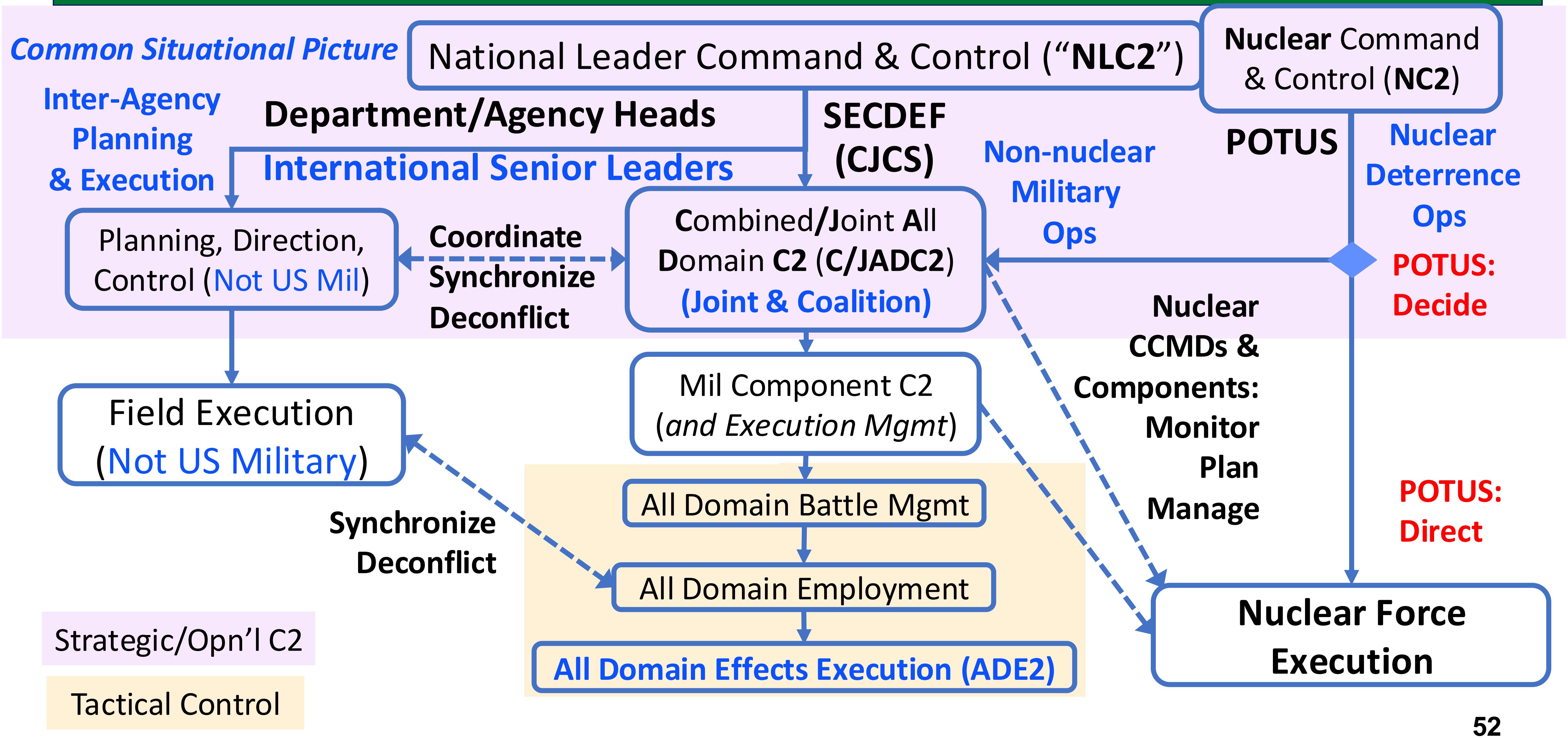


Appropriate evaluation is performed at each stage

"Clean Sheet" Functional Architecture Study



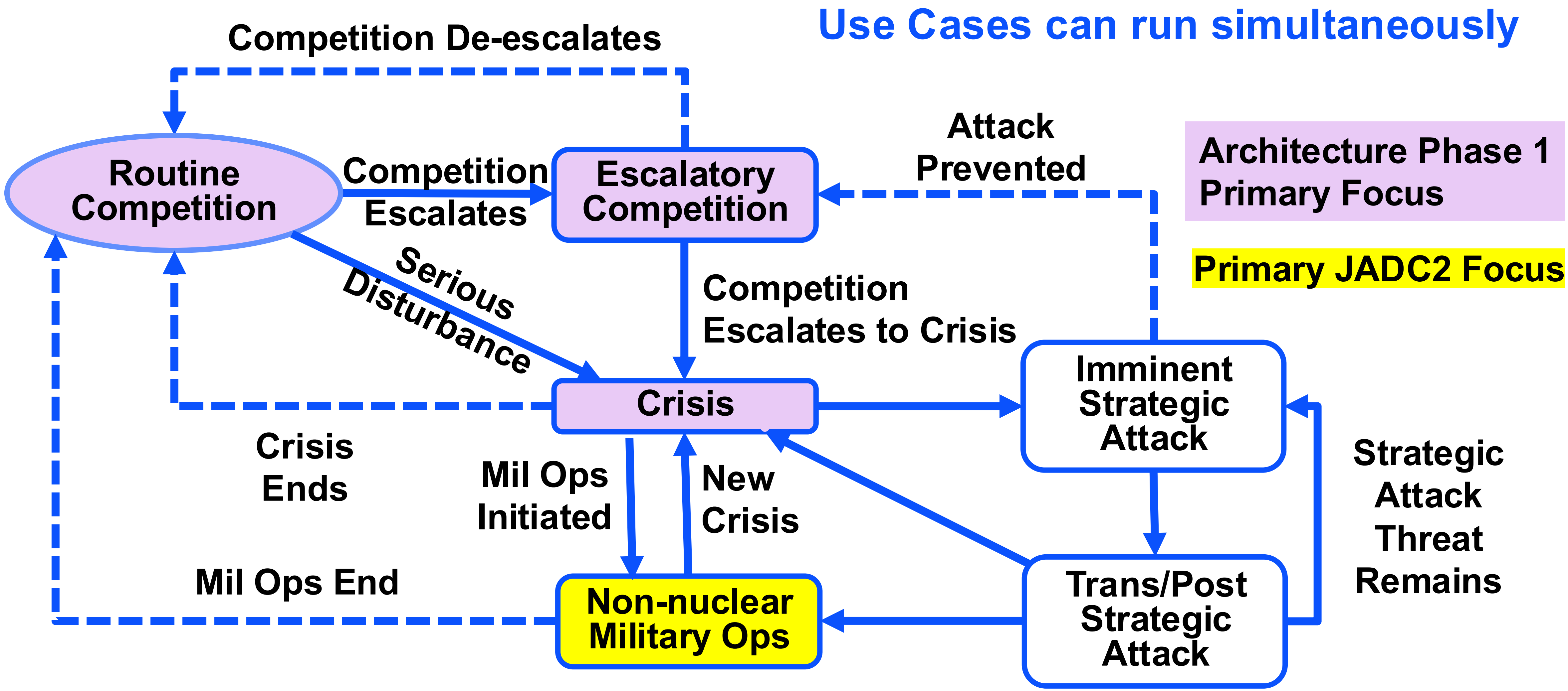
Comprehensive C2 Relationships: NLC2, NC2, C/JADC2, Non-Mil Plan & Execute



Comprehensive C2 Scenarios (and Associated Use Cases)

- Multinational cooperation and capacity building activities: Routine Competition
- Near-peer activities to (apparently) gain competitive advantage (“Long War”): Escalatory
- Near-peer high intensity competition (short of armed conflict): Escalatory competition
- Nuclear-capable adversary regional conflict: Crisis
- Nuclear-capable third-party regional conflict: Crisis
- Near-peer threat to employ nuclear or other strategic weapons: Crisis
- Attack against U.S. allies or partners: Crisis
- Near-peer nuclear weapon employment (not against US): Crisis
- Regional adversary nuclear weapon employment: Crisis
- Nuclear weapon employment in regional conflict: Crisis
- Non-nuclear conflict (Primary JADC2 Focus): Non-nuclear Military Operations
- Actions to degrade U.S. or partner strategic capabilities: Strategic Attack
- Attack against U.S. homeland or territories: Strategic Attack
- Post-nuclear use stability restoration and further use deterrence: Trans/Post Nuclear Use

Use Cases & Relationships



Example CCDR Use Case Description: Crisis

Use Case Element	Description
Goal/Context	Prepare forces, plan & execute response, terminate crisis on favorable terms
Primary Actor & Level	**CCDRs/Strategic**
Stakeholders & Interests	Secretaries of Defense, CJCS, Agency Heads
Scope (Notional System)	**All-domain, Multi-agency mission analysis, strategic & military COA development & assessment & selection, COA planning, plan execution**
Pre-condition(s)	Environmental conditions destabilized
Success End Condition	**Crisis terminated on terms favorable to U.S.**
Min Guarantee (Fail condition)	Crisis terminated on favorable terms but with reduced competitive advantage
Trigger Event	Disturbance escalates to pose threat to U.S. or partner interests
Main Success Scenario	**(Actor-system interactions outlined in CAMEO model)**
Scenario Extensions	Non-nuclear ops escalation, strategic attack
Scenario Variations	Natural disaster, homeland threat, Insider threat, economic threat
Secondary Actors	CCMD Staff, Fed Agency staffs

Capabilities Identified for Prototyping and Experimentation (Selected)

- **Process Data from Multiple Databases**
- **Process Change Detection Algorithms**
- **Detect environmental changes over time**
- **Detect and forecast disturbances**
- **Validate data using alternate sources**
- **Identify information ambiguities and gaps**
- **Leverage historical data to reduce ambiguities**
- **Detect or forecast potential threats**
- **Assess environment during COA execution**
- **Identify connected C2 nodes and status**

Capabilities identified from Main Success Scenario user interactions with a notional functional system

Ontology standards have contributed to significant capability advances

Benefits of DoD Ontology Standardization